— SYDT 5/2 Assignment 3

Part A

x_1	x_2
-2	-1
-1	0
0	1
1	2
2	3

Q1: ① Center the dataset

$$\mu_1 = \frac{(-2)+(-1)+0+1+2}{5} = 0$$

$$\mu_2 = \frac{(-1)+0+1+2+3}{5} = 1$$

mean vector $\mu = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$x_1 - \mu_1$	$x_2 - \mu_2$
$-2-0 = -2$	$-1-1 = -2$
$-1-0 = -1$	$0-1 = -1$
$0-0 = 0$	$1-1 = 0$
$1-0 = 1$	$2-1 = 1$
$2-0 = 2$	$3-1 = 2$

Centered dataset

$$X_c = \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

② Covariance matrix

$$C = \frac{1}{N-1} X_c^T X_c$$

$$= \frac{1}{5-1} \begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \end{bmatrix}$$

③ Eigenvalues & Eigenvectors

$$(C-\lambda I)w = 0 = \begin{bmatrix} 2.5-\lambda & 2.5 \\ 2.5 & 2.5-\lambda \end{bmatrix}$$

$$\det(C-\lambda I) = 0 = (2.5-\lambda)^2 - (2.5)^2$$

$$0 = 6.25 - 5\lambda + \lambda^2 - 6.25$$

$$0 = \lambda^2 - 5\lambda$$

$$0 = \lambda(\lambda - 5)$$

Eigenvalues : $\lambda_1 = 5 \quad \lambda_2 = 0$

Continuous Eigenvectors

$\lambda_1 = 5$ $\begin{bmatrix} 2.5-5 & 2.5 \\ 2.5 & 2.5-5 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = 0$

$-2.5V_1 + 2.5V_2 = 0$

$2.5V_1 - 2.5V_2 = 0$

$V_1 = V_2$

$V_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = 0$ $\begin{bmatrix} 2.5-0 & 2.5 \\ 2.5 & 2.5-0 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = 0$

$2.5V_1 + 2.5V_2 = 0$

$V_1 = -V_2$

$V_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\| V_1 \|_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$

$w_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$\| V_2 \|_2 = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$

$w_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

First principal component (largest eigenvalue): $w_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

Q2: point $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$

① center point: $z = x - \mu = \begin{bmatrix} 3-0 \\ 3-1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

② Project onto PC1: $a = w_1^T z$

$a = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$= \frac{3}{\sqrt{2}} + \frac{2}{\sqrt{2}}$

$= \frac{5}{\sqrt{2}}$

③ Reconstruct point: $\hat{z} = a w_1$

$= \frac{5}{\sqrt{2}}\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$= \begin{bmatrix} 2.5 \\ 2.5 \end{bmatrix}$

④ Add mean: $\hat{x} = \hat{z} + \mu$

$= \begin{bmatrix} 2.5 \\ 2.5 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 2.5 \\ 3.5 \end{bmatrix}$

⑤ L_2 reconstruction error: $E = \| x - \hat{x} \|_2$

$= \left\| \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 2.5 \\ 3.5 \end{bmatrix} \right\|_2$

$= \left\| \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix} \right\|_2$

$= \sqrt{(0.5)^2 + (-0.5)^2}$

$= \frac{1}{\sqrt{2}}$

Q3 point $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

① center point : $z = x - \mu = \begin{bmatrix} 3-0 \\ 4-1 \end{bmatrix} = \boxed{\begin{bmatrix} 3 \\ 3 \end{bmatrix}}$

② project onto PC1 : $a = w_1^T z$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$= \frac{3}{\sqrt{2}} + \frac{3}{\sqrt{2}}$$

$$= \boxed{3\sqrt{2}}$$

③ reconstruct point : $\hat{z} = a w_1$

$$= 3\sqrt{2} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 3 \\ 3 \end{bmatrix}}$$

④ Add mean : $\hat{x} = \hat{z} + \mu$

$$= \begin{bmatrix} 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}$$

⑤ L_2 reconstruction error : $E = \| x - \hat{x} \|_2$

$$= \left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\|_2$$

$$= \left\| \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\|_2$$

$$= \sqrt{0^2 + 0^2}$$

$$= \boxed{0}$$